

CLM/Bristol Babcock Parameter Reference Dictionary

LK09-140

**Communication
Link Module**

***CLM/Bristol Babcock
Parameter Reference
Dictionary***

**LK09-140
4/94**

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About This Publication

This publication is a reference for process engineers, control system engineers, and application engineers who design and implement the Honeywell Communication Link Module on an Application Module of the TDC 3000^X Network. This publication provides reference information about parameters, a dictionary of filenames and explanations, and summary information on CL programs and data segments.

This publication is part of a set of publications that provide parameter reference information for nodes on the LCN. The other members of this set are:

- *Application Module Parameter Reference Dictionary* in the *Implementation/AM* - 1 binder
 - *Process Manager Parameter Reference Dictionary* in the *Implementation/Process Manager* - 2 binder
 - *Logic Manager Parameter Reference Dictionary* in the *Implementation/LM* - 2 binder
 - *Hiway Gateway Parameter Reference Dictionary* in the *Implementation/Hiway Gateway* - 1 binder
 - *Programmable Logic Controller Gateway Parameter Reference Dictionary* in the *Implementation/PLC Gateway* binder
 - *Computer Gateway Parameter Reference Dictionary* in the *Implementation/Computer Gateway* binder
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References

Configuring the AM

Publication Title	Publication Number	Binder Title	Binder Number
<i>Application Module Configuration Forms</i>	AM88-300	Implementation/Configuration Forms	TDC 1031
<i>Application Module Form Instructions</i>	AM12-300	Implementation/AM - 1	TDC 1035-1
<i>Application Module Control Functions</i>	AM09-302	Implementation/AM - 1	TDC 1035-1
<i>Application Module Algorithm Engineering Data</i>	AM09-301	Implementation/AM - 1	TDC 1035-1
<i>Engineer's Reference Manual</i>	SW09-305	Implementation/Startup & Reconfiguration - 2	TDC 1030-2
<i>Data Entity Builder Manual</i>	SW11-311	Implementation/Engineering Operations - 1	TDC 1032

Programming in CL

Publication Title	Publication Number	Binder Title	Binder Number
<i>Control Language/Application Module Overview</i>	SW27-300	Implementation/AM - 2	TDC 1035-2
<i>Control Language/Application Module Reference Manual</i>	AM27-310	Implementation/AM - 2	TDC 1035-2
<i>Control Language/Application Module Data Entry</i>	AM11-385	Implementation/AM - 2	TDC 1035-2

Section 1 – Introduction

1.1 General

This section

This section covers the following topics.

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What this manual is

This publication contains reference information on the TDC 3000^X System parameters that are associated with Honeywell's Communication Link Module (CLM).

The reference information includes three types of data:

- a dictionary that includes all the parameter names associated with the CLM; for each parameter name listed, related data is given (such as a description, default value, and permissible range of values)
 - a dictionary of file names supplied in the release
 - a summary of the names and descriptions of the CL programs and Custom Data Segments in the CLM package
-

1.2 Use of This Publication

When to use it

This publication should be used for reference when:

- building AM data points
 - planning and writing CL programs associated with the AM
 - parameter names appear on the Universal Station displays when the system is in operation.
-

Section 2 – Parameter Dictionary

2.1 General

This section

This section consists of a dictionary that includes all the parameter names associated with the CLM; for each parameter name listed, related data is given (such as a description, default value, and permissible range of values).

This section covers the following topics.

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2.2 Parameters

Definition

A parameter name is a capitalized character string of one to eight alphabetic or alphanumeric characters.

A parameter is a mnemonic which identifies a value used in the system.

2.3 Format

Overview

The standard form of each parameter entry is presented in the format example below.

Source is the name of the CDS module or AM in which the parameter is defined.

Entry format

Point Type AM Regulatory

Source **Default** 0 **Access** **Lock** Engineer

Type Real

Range >= 0.0

2.4 Parameter Name Listing

ACTION

Point Type AM Custom, Builder Point (BUILD_nn)

Builder Action Code

Source BUILD_CDS.CL **Default** 0 **Access** **Lock** Engineer

Type Real

Range >= 0.0

ADJ_PRES

Point Type AM Regulatory

Average barometric pressure, psi

Source B_AGA3.CL **Default** 0 **Access** **Lock** Engineer

Type Real

Range >= 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

ALMOPT	Point Type AM Switch		
	Source	DIG3CDS.CL DIG4CDS.CL DIG5CDS.CL	Default 0 Access Lock Engineer, build visible
	Type	Real	
	Range	0=none 1=offnorml 2=cmddis	

BASE_D	Point Type AM Regulatory		
	Contract base density		
	Source	B_AGA7.CL	Default 0 Access Lock Engineer
	Type	Real	
BASE_P	Range	>= 0.0	

BASE_P	Point Type AM Regulatory		
	Base or contract pressure, psi		
	Source	B_AGA3.CL B_AGA5.CL B_AGA7.CL	Default 0 Access Lock Engineer
	Type	Real	
BASE_T	Range	>= 0.0	

BASE_T	Point Type AM Regulatory		
	Base temperature, degf		
	Source	B_AGA3.CL B_AGA5.CL B_AGA7.CL	Default 0 Access Lock Engineer
	Type	Real	
BASE_T	Range	>= 0.0	

Continued on next page

2.4 Parameter Name Listing, Continued

BREAK

Point Type AM Custom, Builder Point (BUILD_nn)

Interrupt File Input/BackBuild

Source BUILDDCDS.CL **Default** OFF **Access Lock** Engineer

Type Logical

Range OFF, ON

BUILDER

Point Type AM Custom, Builder Point (BUILD_nn)

Builder display invoked

Source BUILDDCDS.CL **Default** OFF **Access Lock** Engineer

Type Logical

Range OFF, ON

C

Point Type AM Regulatory

Calibration factor

Source B_AGA7.CL **Default** NaN **Access Lock** Engineer

Type Real

Range ≥ 0.0

CFGSTAT

Point Type AM Regulatory

Configuration status string from transmitter where:

- I=Invalid
- M=Mismatch

Source B_HWSTI.CL **Default** spaces **Access Lock** Engineer

Type String-32

Continued on next page

2.4 Parameter Name Listing, Continued

CHANNEL

Point Type AM Regulatory

Channel on STI card (not supported by BSAP)

Source B_HWSTI.CL **Default** NaN **Access Lock** Engineer

Type Real

Range ≥ 0.0

CJTACT

Point Type AM Regulatory

STT cold junction. Defines whether the Smart Temperature Transmitter's internal cold junction reference is to be used, or an externally provided cold junction reference is to be used. When the RTU database and the Smart Transmitter databases have different values, MISMATCH(1) will be ON.

Source B_HWSTI.CL **Default** OFF **Access Lock** Engineer

Type LOGICAL

Range OFF Use external cold junction reference.
ON Internal cold junction reference temperature is active.

CKPTCYCL

Point Type AM Custom, Network Point (CLM_nn)

Checkpoint cycle period

Source NETCDS.CL **Default** 4.0 **Access Lock** Engineer

Type Real

Range ≥ 0.0

CKPTSTRT

Point Type AM Custom, Network Point (CLM_nn)

Checkpoint start offset

Source NETCDS.CL **Default** 30.0 **Access Lock** Engineer

Type Real

Range ≥ 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

CLEARALM	Point Type AM Custom, Network Point (CLM_nn) CLM Alarm clear indicator Source NETCDS.CL Default OFF Access Lock Engineer Type Logical Range OFF, ON
CLMFAIL	Point Type AM Custom, Network Point (CLM_nn) Composite CLM failure indicator Source NETCDS.CL Default 0 Access Lock Engineer Type REAL Range >= 0
CLMSTAT	Point Type AM Custom, Network Point (CLM_nn) Composite CLM Status Source NETCDS.CL Default 4 Access Lock Engineer Type Real Range 0 to 4 (GOOD/ERROR/FAILED/UNKNOWN)
CLM_DEV	Point Type AM Regulatory or AM Switch NDT Device Name saved by the Builder Source CLM_DMD3.CL Default spaces Access Lock Engineer Type String

Continued on next page

2.4 Parameter Name Listing, Continued

CLM_DISP	Point Type AM Regulatory STI Custom Display is operating on this tag. On the CDS page (class general) Source B_HWSTI.CL Default OFF Access Lock Engineer Type Logical Range OFF, ON
CLM_OUT	Point Type AM Regulatory or AM Switch Storing to CLM_OUT causes PITE outputs to process Source CLM_DMD3.CL Default OFF Access Lock Engineer Type Logical Range OFF, ON
CLM_PIT	Point Type AM Regulatory or AM Switch NDT PIT table Name saved by the Builder Source CLM_DMD3.CL Default spaces Access Lock Engineer Type String
CLM_PORT	Point Type AM Regulatory or AM Switch NDT Port Device Name saved by the Builder Source CLM_DMD3.CL Default spaces Access Lock Engineer Type String
CLM_PPS	Point Type AM Regulatory or AM Switch Storing to CLM_PPS causes PITE to process Source CLM_DMD3.CL Default OFF Access Lock Engineer Type Logical Range OFF, ON

Continued on next page

2.4 Parameter Name Listing, Continued

CO2MOLE	Point Type AM Regulatory CO ₂ content mole in % Source B_FPV.CL Default 0 Access Lock Engineer Type Real Range >= 0.0
COLFLG()	Point Type AM Regulatory Buffer Collection Flag Source NETCDS.CL Default 0.0 Access Lock Operator Type Real Array (1..4) Range >= 0.0
COMERRS	Point Type AM Regulatory Count of transmitter communication errors detected by HWSTI. The counter is modulo 256. Source B_HWSTI.CL Default 0 Access Lock Engineer Type Real Range 0 to 255
COMMAND	See SCOMMAND .
COUNT	Point Type AM Regulatory Accumulated count Source B_COUNT.CL Default 0 Access Lock Engineer Type Real Range >= 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

CPPTCLM

Point Type AM Custom, Network Point (CLM_nn)

Checkpoint this CLM

Source NETCDS.CL **Default** OFF **Access Lock** Operator

Type Logical

Range OFF, ON

CV

Point Type AM Regulatory

The result of control algorithm processing prior to output processing

Source AM **Default** 0 **Access Lock** Program

Type Real

Range Any real number

CVIN

Point Type AM Regulatory

Output from RTU (class general)

Source B_PID3TM.CL **Default** 0 **Access Lock** Engineer

Type Real

Range ≥ 0.0

C_SPAN

Point Type AM Regulatory

Multiplier

Source B_COUNT.CL **Default** 0 **Access Lock** Engineer

Type Real

Range ≥ 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

C_ZERO

Point Type AM Regulatory

Scale count to 0% value (defaults to 0)

Source B_COUNT.CL **Default** 0 **Access Lock** Engineer

Type Real

Range >= 0.0

D1

Point Type AM Switch

Input 1 status

Source B_DG3CDS.CL **Default** 0 **Access Lock** Engineer
B_DG4CDS.CL
B_DG5CDS.CL

Type Logical

Range OFF, ON

D1_0

Point Type AM Switch

D1_Zero PV State (NODINPTS=1)

Source DIG3CDS.CL **Default** \$ord0 **Access Lock** Engineer,
DIG4CDS.CL build visible
DIG5CDS.CL

Type Self-Defining Enumeration

Range \$ord0, \$ord1, \$ord2, \$ord3, \$ord4

Note: For 0 input conn., \$ord0=S1STATES(0) describes D1=0
\$ord1=S1STATES(1) describes D1=0
\$ord2=S1STATES(2) describes D1=0
\$ord3=S1STATES(3) describes D1=0 otherwise
\$ord4=S1STATES(4) describes D1=0

Note: For 4-state points, \$ord3 must be the user's description of bad PV,
\$ord2 is usually for moving.
For 5-state points, \$ord4 must be the user's description of bad PV,
\$ord3 is usually for moving.

Continued on next page

2.4 Parameter Name Listing, Continued

D1_1

Point Type AM Switch

D1_One PV State (NODINPTS=1)

Source DIG3CDS.CL **Default** \$ord0 **Access Lock** Engineer
DIG4CDS.CL
DIG5CDS.CL

Type Self-Defining Enumeration

Range \$ord0, \$ord1, \$ord2, \$ord3, \$ord4

Note: For 0 input conn., \$ord0=S1STATES(0) describes D1=1
\$ord1=S1STATES(1) describes D1=1
\$ord2=S1STATES(2) describes D1=1
\$ord3=S1STATES(3) describes D1=1 otherwise
\$ord4=S1STATES(4) describes D1=1

Note: For 4-state points, \$ord3 must be the user's description of bad PV,
\$ord2 is usually for moving.
For 5-state points, \$ord4 must be the user's description of bad PV,
\$ord3 is usually for moving.

D2

Point Type AM Switch

Input 2 status

Source B_DG3CDS.CL **Default** OFF **Access Lock** Engineer
B_DG4CDS.CL
B_DG5CDS.CL

Type Logical

Range OFF, ON

Continued on next page

2.4 Parameter Name Listing, Continued

D2D1_00	Point Type AM Switch
	D2_D1 Zero_Zero PV State (NODINPTS=2)
	Source DIG3CDS.CL Default \$ord0 Access Lock Engineer, DIG4CDS.CL build visible DIG5CDS.CL
	Type Self-Defining Enumeration
	Range \$ord0, \$ord1, \$ord2, \$ord3, \$ord4
	Note: For 4-state points, \$ord3 must be the user's description of bad PV, \$ord2 is usually for moving. For 5-state points, \$ord4 must be the user's description of bad PV, \$ord3 is usually for moving.
D2D1_01	Point Type AM Switch
	D2_D1 Zero_One PV State (NODINPTS=2)
	Source DIG3CDS.CL Default \$ord0 Access Lock Engineer, DIG4CDS.CL build visible DIG5CDS.CL
	Type Self-Defining Enumeration
	Range \$ord0, \$ord1, \$ord2, \$ord3, \$ord4
	Note: For 4-state points, \$ord3 must be the user's description of bad PV, \$ord2 is usually for moving. For 5-state points, \$ord4 must be the user's description of bad PV, \$ord3 is usually for moving.

Continued on next page

2.4 Parameter Name Listing, Continued

D2D1_10

Point Type AM Switch

D2_D1 One_Zero PV State (NODINPTS=2)

Source DIG3CDS.CL **Default** \$ord0 **Access Lock** Engineer,
DIG4CDS.CL build visible
DIG5CDS.CL

Type Self-Defining Enumeration

Range \$ord0, \$ord1, \$ord2, \$ord3, \$ord4

Note: For 4-state points, \$ord3 must be the user's description of bad PV,
\$ord2 is usually for moving.
For 5-state points, \$ord4 must be the user's description of bad PV,
\$ord3 is usually for moving.

D2D1_11

Point Type AM Switch

D2_D1 One_One PV State (NODINPTS=2)

Source DIG3CDS.CL **Default** \$ord0 **Access Lock** Engineer,
DIG4CDS.CL build visible
DIG5CDS.CL

Type Self-Defining Enumeration

Range \$ord0, \$ord1, \$ord2, \$ord3, \$ord4

Note: For 4-state points, \$ord3 must be the user's description of bad PV,
\$ord2 is usually for moving.
For 5-state points, \$ord4 must be the user's description of bad PV,
\$ord3 is usually for moving.

Continued on next page

2.4 Parameter Name Listing, Continued

DAMPING

Point Type AM Regulatory

Damping index defines the first order PV filtering option for the Smart Transmitter. Although a user could implement PV filtering in the AM with the TF parameter, DAMPING is the preferred method. When the RTU database and the Smart Transmitter databases have different values, MISMATCH(4) will be ON.

Source B_HWSTI.CL **Default** 0 **Access Lock** Supervisor

Type Real

Range 0 to 9

Transmitter Type			
	SPT	STT	SFM
0	0.0	0.0	0.0
1	0.16	0.30	0.5
2	0.32	0.70	1.0
3	0.48	1.5	2.0
4	1.0	3.1	3.0
5	2.0	6.3	4.0
6	4.0	12.7	5.0
7	8.0	25.5	10.0
8	16.0	51.1	50.0
9	32.0	102.3	100.0

DATABUF

Point Type AM Custom, Builder Point (BUILD_nn)

Miscellaneous Data Usage array

Source BLD2CDS.CL **Default** 0 **Access Lock** Engineer

Type Real ARRAY(1..20)

Range ≥ 0.0

DEADBAND

Point Type AM Regulatory

Output deadband

Source B_PID3TM.CL **Default** 0 **Access Lock** Engineer

Type Real

Range ≥ 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

DEBUGLEV

Point Type AM Custom, Builder and Network Points (BUILD_nn and CLM_nn)

Debug Level for program tracing. Messages are displayed based on this value > 5 (Process tracing and status check).

Source BUILD_CDS.CL **Default** 0 **Access Lock** Engineer
NET_CDS.CL

Type Real

Range >= 0.0

DECONF

Point Type AM Regulatory

Digitally Enhanced protocol configured transmission mode. When the RTU database and the Smart Transmitter databases have different values, MISMATCH(3) will be ON.

Source B_HWSTI.CL **Default** 4 **Access Lock** Engineer,
build visible

Type Real

Range 1 to 4

Definition of Bristol DE Configuration Mode (DECONF) Signal Codes

- | | |
|------------------------------------|------------|
| 1 Process Variable (PV) Only | (PV) |
| 2 PV and Secondary Variable (SV) | (PV_SV) |
| 3 PV and transmitter database | (PV_DB) |
| 4 PV, SV, and transmitter database | (PV_SV_DB) |
-

DEFARR

Point Type AM Custom, Builder Point (BUILD_nn)

Definition Data Array

Source BUILD_CDS.CL **Default** spaces **Access Lock** Engineer

Type ARRAY(1..50) of String

Continued on next page

2.4 Parameter Name Listing, Continued

DEFNUM	Point Type AM Custom, Builder Point (BUILD_nn)
	Number of definitions this entry
	Source BUILD_CDS.CL Default 0 Access Lock Engineer
	Type Real Range >= 0.0
DELAY	Point Type AM Switch
	Provides a time delay to allow the relays to settle after an output.
	Source B_CBO.CL Default NaN Access Lock Engineer
	Type Real Range >= 0
DEMNDIN1, DEMNDIN2, DEMNDIN3, DEMNDIN4	Point Type AM Custom, Network Point (CLM_nn)
	Buffer # (1–4) demand address index #
	Source NETCDS1.CL (DEMNDIN1) Default 0 Access Lock Operator NETCDS2.CL (DEMNDIN2) NETCDS3.CL (DEMNDIN3) NETCDS4.CL (DEMNDIN4)
	Type Real Range >= 0.0
DENS_SW	Point Type AM Regulatory
	Select densitometer or gravitometer
	Source B_AGA7.CL Default OFF Access Lock Engineer
	Type Logical Range OFF, ON
DETSTAT	Point Type AM Custom, Builder Point (BUILD_nn)
	Error Detailed Status
	Source BUILD_CDS.CL Default 0 Access Lock Engineer
	Type Real Range >= 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

DEVICE

Point Type AM Regulatory or AM Switch

Card cage slot number. Transfer of this data out of the RTU is not supported by BSAP. The user may enter a value here for operator reference.

Source B_ANIN.CL **Default** 0 **Access Lock** Engineer
 B_ANOUT.CL
 B_CBO.CL
 B_COUNT.CL
 B_DIGIN.CL
 B_HWSTI.CL

Type Real

Range Any real value

**DEVICE1,
DEVICE2**

Point Type AM Switch

D1 (D2) signal, input slot number. Transfer of this data out of the RTU is not supported by BSAP. The user may enter a value here for operator reference.

Source B_DG3CDS.CL **Default** 0 **Access Lock** Engineer
 B_DG4CDS.CL
 B_DG5CDS.CL

Type Real

Range Any real value

**DEVICE01,
DEVICE02,
DEVICE03**

Point Type AM Switch

OP1 (OP2, OP3) signal, output slot number. Transfer of this data out of the RTU is not supported by BSAP. The user may enter a value here for operator reference.

Source B_DG3CDS.CL **Default** 0 **Access Lock** Engineer
 B_DG4CDS.CL
 B_DG5CDS.CL

Type Real

Range Any real value

Continued on next page

2.4 Parameter Name Listing, Continued

DIFFPRES	Point Type AM Regulatory Measured pressure drop across an orifice plate Source B_AGA3.CL Default 0 Access Lock Engineer Type Real Range >= 0.0
DONE	Point Type AM Regulatory Done Flag. Flag is used with the eight transmitter COMMANDs to indicate completion of a transmitter command: • set to OFF at the beginning of a transmitter command and • set to ON when the command is completed. Source B_HWSTI.CL Default ON Access Lock Engineer Type Logical Range OFF, ON
ENABLE	Point Type AM Switch Selectively enables the updating of the output terminal. Source B_CBO.CL Default OFF Access Lock Engineer Type Logical Range ON, OFF
ENRG_CNV	Point Type AM Regulatory Energy conversion factor Source B_AGA5.CL Default 0 Access Lock Engineer Type Real Range >= 0.0
ENTARR()	Point Type AM Custom, Builder Point (BUILD_nn) Entry Type Array Source BLD2CDS.CL Default 0 Access Lock Engineer Type ARRAY(1..120) of String

Continued on next page

2.4 Parameter Name Listing, Continued

ENTRY	Point Type AM Custom, Builder Point (BUILD_nn)		
	Entry		
	Source BUILDCDS.CL	Default 0	Access Lock Engineer
	Type Real		
	Range 0=none,1=NDT,2=PIT,3=PITE,4=PITG		
ENTRYNAM	Point Type AM Custom, Builder Point (BUILD_nn)		
	Entry Name		
	Source BUILDCDS.CL	Default spaces	Access Lock Engineer
	Type String		
ENTRYTYP	Point Type AM Custom, Builder Point (BUILD_nn)		
	Entry Type (TDF NAME)		
	Source BUILDCDS.CL	Default spaces	Access Lock Engineer
	Type String		
FB1, FB2, FB3	Point Type AM Switch		
	Output 1 (2,3) Signals as read back from the RTU reset terminals.		
	Source DIG3CDS.CL DIG4CDS.CL DIG5CDS.CL	Default OFF	Access Lock Program
	Type Logical		
	Range OFF, On		

Continued on next page

2.4 Parameter Name Listing, Continued

FBTIME

Point Type AM Switch

Feedback time

Source B_DG3CDS.CL **Default** NaN **Access Lock** Engineer,
B_DG4CDS.CL build visible
B_DG5CDS.CL

Type Real

Range ≥ 0.0

FLOW

Point Type AM Regulatory

Corrected gas flow rate, MSCFH

Source B_AGA3.CL **Default** NaN **Access Lock** Engineer

Type Real

Range ≥ 0.0

FLOW_D

Point Type AM Regulatory

Density of flowing gas as measured by densitometer

Source B_AGA7.CL **Default** NaN **Access Lock** Engineer

Type Real

Range ≥ 0.0

FLOW_P

Point Type AM Regulatory

Flowing gas pressure, psia

Source B_AGA7.CL **Default** NaN **Access Lock** Engineer

Type Real

Range ≥ 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

FLOW_SW	Point Type AM Regulatory
	Select equation
Source	B_AGA7.CL Default OFF Access Lock Engineer
Type	Logical
Range	OFF, ON

FLOW_T	Point Type AM Regulatory
	Flowing temperature of gas, degf
Source	B_AGA3.CL Default NaN Access Lock Engineer B_AGA7.CL B_FPV.CL
Type	Real
Range	>= 0.0

FMTARR()	Point Type AM Custom, Builder Point (BUILD_nn)
	Format Data Array
Source	BUILDCDS.CL Default spaces Access Lock Engineer
Type	ARRAY(1..50) of String

FPV_IN	Point Type AM Regulatory
	Supercompressibility factor of gas
Source	B_AGA3.CL Default NaN Access Lock Engineer B_AGA5.CL B_AGA7.CL
Type	Real
Range	>= 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

FREQ

Point Type AM Regulatory

Frequency of the input signal

Source B_COUNT.CL **Default** NaN **Access Lock** Engineer

Type Real

Range ≥ 0.0

FREQ6050

Point Type AM Regulatory

STT Line Frequency Filter. Defines the 60/50 Hz line frequency filter configuration needed for an STI Temperature Transmitter. When the RTU database and the Smart Transmitter databases have different values, MISMATCH(1) will be ON.

Source B_HWSTI.CL **Default** OFF **Access Lock** Engineer

Type Logical

Range OFF – 60 Hz
ON – 50 Hz

F_SPAN

Point Type AM Regulatory

Frequency multiplier

Source B_COUNT.CL **Default** NaN **Access Lock** Engineer

Type Real

Range ≥ 0.0

F_ZERO

Point Type AM Regulatory

0% value of frequency

Source B_COUNT.CL **Default** NaN **Access Lock** Engineer

Type Real

Range ≥ 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

GENATTR()

Point Type AM Custom

Display attributes

Source GENRLCDS.CL **Default** 6 **Access Lock** Engineer

Type ARRAY (1..40) of NUMBER

Range 1 to 8

CODE	COLOR	BLINK	REVERSE	COLOR & TARGET
1	RED	11	21	101
2	YELLOW	12	22	102
3	GREEN	13	23	103
4	MAGENTA	14	24	104
5	BLUE	15	25	105
6	CYAN	16	26	106
7	WHITE	17	27	107
8	BLACK	NO	NO	108

GENCOM

Point Type AM Custom

Display command

Source GENRLCDS.CL **Default** 0 **Access Lock** Engineer

Type NUMBER

Range >= 0

GENDISP()

Point Type AM Custom

Display strings

Source GENRLCDS.CL **Default** spaces **Access Lock** Engineer

Type NUMBER

Range ARRAY (1..40) of String

GENNODE

Point Type AM Custom

Display node identification

Source GENRLCDS.CL **Default** **Access Lock** Engineer

Type NUMBER

Range >= 0

Continued on next page

2.4 Parameter Name Listing, Continued

GENSTAT	Point Type AM Custom Display status Source GENRLCDS.CL Default 0 Access Lock Engineer Type NUMBER Range >= 0
GENSTATE	Point Type AM Custom Display state Source GENRLCDS.CL Default 0 Access Lock Engineer Type NUMBER Range >= 0
GENSUB	Point Type AM Custom Display subpicture used Source GENRLCDS.CL Default 0 Access Lock Engineer Type NUMBER Range 1 to 3 1 = 16 character strings/120 per display/40 from each LCN point 2 = 20 character strings/96 per display/32 from each LCN point 3 = 40 character strings/48 per display/16 from each LCN point
GENTAR1	Point Type AM Custom Target string 1 Source GENRLCDS.CL Default " APPL " Access Lock Engineer Type STRING
GENTARZ	Point Type AM Custom Target string 2 Source GENRLCDS.CL Default "SCHEMATIC" Access Lock Engineer Type STRING

Continued on next page

2.4 Parameter Name Listing, Continued

GRAV_P	Point Type AM Regulatory		
	Pressure at gravitometer		
	Source B_AGA7.CL	Default NaN	Access Lock Engineer
	Type Real		
	Range >= 0.0		

GRAV_T	Point Type AM Regulatory		
	Temperature at gravitometer		
	Source B_AGA7.CL	Default NaN	Access Lock Engineer
	Type Real		
	Range >= 0.0		

GROUPIN1, GROUPIN2, GROUPIN3, GROUPIN4	Point Type AM Custom, Network Point (CLM_nn)		
	Buffer # (1–4) group address index #		
	Source NETCDS1.CL (GROUPIN1) Default 0 Access Lock		
	Operator		
	NETCDS2.CL (GROUPIN2)		
	NETCDS3.CL (GROUPIN3)		
	NETCDS4.CL (GROUPIN4)		
	Type Real		
	Range >= 0.0		

INDXNO()	Point Type AM Custom, Network Point (CLM_nn)		
	Buffer index number		
	Source NETCDS.CL	Default 0	Access Lock Operator
	Type REAL ARRAY (1..4)		
	Range >= 0.0		

Continued on next page

2.4 Parameter Name Listing, Continued

INIT1,
INIT2

Point Type AM Switch

D1 (D2) signal, number of DI field wiring terminal: Transfer of this data out of the RTU is not supported by BSAP. The user may enter a value here for operator reference.

Source B_DG3CDS.CL **Default** NaN **Access Lock** Engineer
 B_DG4CDS.CL
 B_DG5CDS.CL

Type Real

Range >= 0.0

INITIAL

Point Type AM Regulatory or AM Switch

Field wiring terminal number: Transfer of this data out of the RTU is not supported by BSAP. The user may enter a value here for operator reference.

Source B_ANIN.CL **Default** NaN **Access Lock** Engineer
 B_ANOUT.CL
 B_COUNT.CL
 B_DIGIN.CL

Type Real

Range Any real number, NaN

INITO1,
INITO2,
INITO3

Point Type AM Switch

OP1 (OP2,OP#) signal, number of DO field wiring terminal

Source B_DG3CMP.CL **Default** NaN **Access Lock** Engineer
 B_DG4CDS.CL
 B_DG5CDS.CL

Type Real

Range Any real number, NaN

Continued on next page

2.4 Parameter Name Listing, Continued

INLINE1,
INLINE2,
INLINE3,
INLINE4

Point Type AM Custom, Network Point (CLM_nn)

Buffer # (1–4) Entries

Source NETCDS1.CL (INLINE1) **Default** 0 **Access Lock** Operator
NETCDS2.CL (INLINE2)
NETCDS3.CL (INLINE3)
NETCDS4.CL (INLINE4)

Type Real

Range ≥ 0.0

INPFILE

Point Type AM Custom, Builder Point (BUILD_nn)

Input File Pathname (User Linkage File)

Source BUILD_CDS.CL **Default** spaces **Access Lock** Engineer

Type String

INPUT

Point Type AM Regulatory

Measured variable

Source B_ANIN.CL **Default** 0 **Access Lock** Engineer,
B_PID3TM.CL class general
B_HWSTI.CL

Type Real

Range ≥ 0.0

K

Point Type AM Regulatory

Adjust output to desired units (AGA7) or proportional gain (PID)

Source B_AGA7.CL **Default** 0 **Access Lock** Engineer
B_PID3TM.CL

Type Real

Range ≥ 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

LNCTR()	Point Type AM Custom, Network Point (CLM_nn) Work Area Number Source NETCDS.CL Default (1.0,2.0,3.0,4.0,5.0,6.0,7.0,8.0,9.0,10.0,11.0,12.0,13.0,14.0,15.0) Access Lock Engineer Type Real ARRAY (1..15) Range 1 – 15
LOCK	Point Type AM Custom, Builder Point (BUILD_nn) Lock Display while processing Source BUILD_CDS.CL Default OFF Access Lock Engineer Type Logical Range OFF, ON
LRL	Point Type AM Regulatory Lower Range Limit: Indicates the lower range limit of the PV at the Smart Transmitter. This limit is a fixed limit and cannot be changed. Refer to the description of the STI_EU parameter for the LRL engineering units. Source B_HWSTI.CL Default 0 Access Lock Engineer Type Real Range N/A
LRV	Point Type AM Regulatory Lower Range Value: Defines the lower end of the operating range for the PV parameter value. Refer to the STI_EU parameter for the LRV engineering units. User entry for PVEULO is the user-entered engineering unit value that corresponds to LRV. Valid values for LRV must be in the range: LRL => (LRV VALUE) =< URV. A value of NaN is invalid. When the RTU and the Smart Transmitter databases have different values, MISMATCH(10) will be ON. This parameter is mapped as PVEULO. Source B_HWSTI.CL Default 0 Access Lock Supervisor Type Real Range N/A

Continued on next page

2.4 Parameter Name Listing, Continued

MISMATCH()

Point Type AM Regulatory

Mismatch Array: If MISMATCH(1)=ON, at least one configuration parameter in the Smart Transmitter does not match the signal value representing the parameter.

If MISMATCH(1)=OFF, no parameters mismatch. (Invalid configuration parameters are treated as an implicit mismatch, since it is not possible for the transmitter to have an invalid parameter value.)

Mismatch detection occurs only if the transmitter's DECONF parameter is equal to PV_DB or PV_SV_DB. A mismatch condition is assumed when the system first powers up, and when an SFC writes to a parameter that affects the PV. The Questionable Data status is set for all signals with a mismatch condition. The mismatch condition can be cleared by a Download or Upload command, or when the transmitter sends its database and all parameters are compared equal.

In the following, numeric signals are checked for a mismatch with the transmitter. The string and logical signals (–) are not checked.

Definition of Bristol Babcock MISMATCH LOGICAL ARRAY positions

- 1 any fault
- 2 SENSRTYP
- 3 DECONF
- 4 DAMPING
- 5 PVCHAR
- 6 CJTACT* (MISMATCH is not supported in Bristol Babcock RTU for LOGICAL signals)
- 7 PIUOTDCF* (MISMATCH is not supported in Bristol Babcock RTU for LOGICAL signals)
- 8 STITAG* (MISMATCH is not supported in Bristol Babcock RTU for STRING signals)
- 9 FREQ6050* (MISMATCH is not supported in Bristol Babcock RTU for LOGICAL signals)
- 10 LRV
- 11 URV
- 12 URL

Source B_HWSTI.CL **Default** OFF

Access Lock Engineer, on CDS page (class general)

Type LOGICAL ARRAY (1..12)

Range OFF, ON

Continued on next page

2.4 Parameter Name Listing, Continued

MODATTR	Point Type AM Regulatory Mode attribute for control algorithms Source AM Default OPERATOR Access Lock Operator Type MODATTR enumeration Range NONE, OPERATOR, PROGRAM
MODE	Point Type AM Regulatory Current mode for a control algorithm Source AM Default 0 Access Lock Engineer Type MODE Enumeration Range MAN, AUTO, CAS, NORMAL
NBUFFER()	Point Type AM Custom, Network Point (CLM_nn) NDT buffer flag Source NETCDS.CL Default 0 Access Lock Operator Type Real ARRAY (1..4) Range >= 0.0
NEWSTATE()	Point Type AM Custom, Network Point (CLM_nn) NDT State change flag Source NETCDS.CL Default OFF Access Lock Operator Type LOGICAL ARRAY (1..18) Range OFF, ON
NFAIL()	Point Type AM Custom, Network Point (CLM_nn) NDT failure indicator Source NETCDS.CL Default 0 Access Lock Engineer Type Real ARRAY (1..18) Range >= 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

NMODE

Point Type

AM Regulatory

Normal mode for this data point.

Source

AM

Default

None

Access Lock

Supervisor

Type

MODE enumeration

Range

>= 0.0

NMOLE

Point Type

AM Regulatory

Nitrogen content mole in %

Source

B_FPV.CL

Default

0

Access Lock

Engineer

Type

Real

Range

>= 0.0

NNAME()

Point Type

AM Custom, Network Point (CLM_nn)

NDT PORT etc. name

Source

NETCDS.CL

Default

spaces

Access Lock

Engineer

Type

ARRAY (1..18) of String

NODINPTS

Point Type

AM Switch

Dig. Input Connections (0–2). Two input connections requires at least 4 states.

Source

DIG3CDS.CL
DIG4CDS.CL
DIG5CDS.CL

Default

0

Access Lock

Engineer,
build visible

Type

Real

Range

0, 1, 2

Continued on next page

2.4 Parameter Name Listing, Continued

NODOPTS	Point Type AM Switch # Dig. Output Connections (0–3 <=nostates) Source DIG3CDS.CL Default 0 Access Lock Engineer, DIG4CDS.CL build visible DIG5CDS.CL Type Real Range 0, 1, 2, 3
NSTATE()	Point Type AM Custom, Network Point (CLM_nn) NDT State Source NETCDS.CL Default 0 Access Lock Engineer Type Real ARRAY (1..18) Range >= 0.0
NSTATUS()	Point Type AM Custom, Network Point (CLM_nn) NDT Status Source NETCDS.CL Default 0 Access Lock Engineer Type Real ARRAY (1..18) Range >= 0.0
NUMBR1(), NUMBR2(), NUMBR3(), NUMBR4()	Point Type AM Custom, Network Point (CLM_nn) Buffer # (1–4) Numbers Source NETCDS1.CL (NUMBR1) Default 0 Access Lock Operator NETCDS2.CL (NUMBR2) NETCDS3.CL (NUMBR3) NETCDS4.CL (NUMBR4) Type Real ARRAY (1..300) Range >= 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

OP

Point Type AM Regulatory

The output of a regulatory control algorithm, in %, after control output processing.

Source AM **Default** 0 **Access Lock** Engineer

Type Real

Range >= 0.0

OP1,
OP2,
OP3

Point Type AM Switch

Output 1 (2,3) Signal

Source DIG3CDS.CL **Default** OFF **Access Lock** Engineer
DIG4CDS.CL
DIG5CDS.CL

Type Logical

Range OFF, ON

OPHILM

Point Type AM Regulatory

Output high limit in percent

Source AM **Default** 105.0 **Access Lock** Supervisor

Type Real

Range OPLOLM TO 106.9

OPLOLM

Point Type AM Regulatory

Output low limit in percent

Source AM **Default** -5.0 **Access Lock** Engineer

Type Real

Range -6.9 to OPHILM

Continued on next page

2.4 Parameter Name Listing, Continued

ORIF_D	Point Type AM Regulatory		
	Orifice diameter, inches		
	Source B_AGA3.CL	Default 0	Access Lock Engineer
	Type Real		
	Range >= 0.0		
ORIF_K	Point Type AM Regulatory		
	Orifice constant		
	Source B_AGA3.CL	Default 0	Access Lock Engineer
	Type Real		
	Range >= 0.0		
OUTFILE	Point Type AM Custom, Builder Point (BUILD_nn)		
	Optional Output File Pathname		
	Source BUILDCDS.CL	Default spaces	Access Lock Engineer
	Type String		
OUTFILOP	Point Type AM Custom, Builder Point (BUILD_nn)		
	Output File Option Selected		
	Source BUILDCDS.CL	Default OFF	Access Lock Engineer
	Type Logical		
	Range OFF, ON		
OUTPUT	Point Type AM Regulatory		
	For FPV.CL, computed supercompressibility		
	For AGA7.CL, corrected gas flow rate, MSCFH		
	For AGA5.CL, gas energy (module output)		
	Source B_AGA5.CL	Default 0	Access Lock Engineer, class general
	B_AGA7.CL		
	B_ANOUT.CL		
	B_FPV.CL		
	B_PID3TM.CL		
	Type Real		
Range >= 0.0			

Continued on next page

2.4 Parameter Name Listing, Continued

OUTPUT	Point Type AM Switch
	Control output command to the CBO DO field terminal
	Source B_CBO.CL Default 0FF Access Lock Engineer
	Type Logical Range OFF, ON
PA	Point Type AM Regulatory
	Average barometric pressure, psi
	Source B_AGA7.CL Default 0 Access Lock Engineer
	Type Real Range >= 0.0
PASCALSW	Point Type AM Custom, Network Point (CLM_nn)
	Pascal Run Switch
	Source NETCDS.CL Default ON Access Lock Engineer
	Type Logical Range OFF, ON
PATHIND	Point Type AM Regulatory
	Control path indicator. Indicates whether control information is currently propagated normally, on hold, in initialization, or an override path is selected.
	Source AM Default MAN Access Lock Read Only
	Type PATHIND enumeration Range FWD, HOLD, INIT, OR

Continued on next page

2.4 Parameter Name Listing, Continued

PIPE_D	Point Type AM Regulatory Inside diameter, inches Source B_AGA3.CL Default 0 Access Lock Engineer Type Real Range >= 0.0
PITNAIN1, PITNAIN2, PITNAIN3, PITNAIN4	Point Type AM Custom, Network Point (CLM_nn) Buffer # (1-4) PIT Name address index # Source NETCDS1.CL (PITNAIN1) Default 0 Access Lock Operator NETCDS2.CL (PITNAIN2) NETCDS3.CL (PITNAIN3) NETCDS4.CL (PITNAIN4) Type Real Range >= 0.0
PITNAM	Point Type AM Custom, Builder Point (BUILD_nn) Assoc PIT Name for PITE Source BUILD_CDS.CL Default spaces Access Lock Engineer Type String
PIUOTDCF	Point Type AM Regulatory Open Thermocouple Detection Enable. Defines whether the point is to detect an open thermocouple condition. This parameter is configurable for each STI point. Direction is determined by a wire jumper in the transmitter. When the RTU database and the Smart Transmitter databases have different values, MISMATCH(1) will be ON. Source B_HWSTI.CL Default OFF Access Lock Engineer Type Logical Range OFF disable open T/C detection ON enable open T/C detection

Continued on next page

2.4 Parameter Name Listing, Continued

PORTCD()

Point Type

AM Custom, Network Point (CLM_nn)

Buffer port code

Source

NETCDS.CL

Default

0

Access Lock

Operator

Type

Real ARRAY (1..4)

Range

>= 0.0

PROCNAM

Point Type

AM Custom, Builder Point (BUILD_nn) or Network Point (CLM_nn)

Program/Process Identification

Source

BUILD_CDS.CL
NETCDS.CL

Default

Spaces

Access Lock

Engineer

Type

String

Range

>= 0.0

PROCOPT

Point Type

AM Custom, Builder Point (BUILD_nn)

Process Option for Multiple Build

Source

BUILD_CDS.CL

Default

0

Access Lock

Engineer

Type

Real

Range

>= 0.0

PTPARIN1,
PTPARIN2,
PTPARIN3,
PTPARIN4

Point Type

AM Custom, Network Point (CLM_nn)

Buffer # (1–4) point.par address index #

Source

NETCDS1.CL (PTPARIN1)
NETCDS2.CL (PTPARIN2)
NETCDS3.CL (PTPARIN3)
NETCDS4.CL (PTPARIN4)

Default

0

Access Lock

Operator

Type

Real

Range

>= 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

PVCHAR

Point Type AM Regulatory

Transmitter PV characterization. When the RTU database and the Smart Transmitter databases have different values, MISMATCH(5) will be ON.

Source B_HWSTI.CL **Default** 12 **Access Lock** Engineer,
build visible

Type Real

Range 0 – 23

Definition of STI PV Characterization Codes

	Code	STT	SDP	SGP/SAP	SFM
0	JTHERM	X			
1	KTHERM	X			
2	ETHERM	X			
3	TTHERM	X			
4	BTHERM	X			
5	STHERM	X			
6	RTHERM	X			
7					
8	DINRTD	X			
9	JISRTD	X			
10	NICKLRTD	X			
11					
12	LINEAR		X	X	X
13	SQRROOT		X		
14	NTHERM	X			
15	PT200RTD	X			
16	PT500RTD	X			
17	CU10RTD	X			
18	CU25RTD	X			
19	RHRAD	X			
20	W5W26TC	X			
21	W3W25TC	X			
22	NINIMOTC	X			
23	RTD_Ohms	X			

PVNORMAL

Point Type AM Switch

Contains a character string that represents the position of the normal state of the point in the S1STATES list.

Source DIG3CDS.CL **Default** \$ord0 **Access Lock** Engineer,
DIG4CDS.CL build visible
DIG5CDS.CL

Type \$3ords, self-defining enumeration of S1STATES

Range \$ord0, \$odr1, \$ord2 (3 or more states only)

Continued on next page

2.4 Parameter Name Listing, Continued

PWRFAIL	Point Type AM Regulatory Reserved by Bristol Babcock for future use. Source B_HWSTI.CL Default 0 Access Lock Engineer Type Real Range >= 0.0
RATE	Point Type AM Regulatory Uncorrected gas flow (either rate or volume) Source B_AGA7.CL Default 0 Access Lock Engineer Type Real Range >= 0.0
REFRESHCD()	Point Type AM Custom, Network Point (CLM_nn) Buffer refresh indicator Source NETCDS.CL Default 0 Access Lock Operator Type Real ARRAY (1..4) Range 0 – 7 0 = No refresh 1 = Refresh of Schematic CLMSBUF# status data 2 = Refresh of Schematic CLMSBUF# detail status data 3 = Refresh of Schematic CLMPIT# statistical data 4 = Refresh of Schematic CLMGRP# statistical data 5 = Refresh of Schematic CLMPITE# statistical data 6 = Refresh of Schematic CLMGRP# selected statistical data 7 = Refresh of Schematic CLMPITE# selected statistical data
RELNAME	Point Type AM Custom, Network Point (CLM_nn) Release Control Identification Source NETCDS.CL Default spaces Access Lock Program Type String

Continued on next page

2.4 Parameter Name Listing, Continued

RESET	Point Type AM Regulatory		
	Resets the counter (Counter)		
	Tracks the state of the CBO DO field terminal (CBO)		
	Source B_COUNT.CL B_CBO.CL	Default OFF	Access Lock Engineer
	Type Logical		
	Range OFF, ON		
RESTART	Point Type AM Regulatory or AM Switch		
	Restart Type.		
	Note: This is a reserved name that must be represented in CL as RESTART.		
	Source AM	Default NONE	Access Lock Engineer
	Type RESTART enumeration		
	Range NONE, HOT, WARM, COLD, PTACTVN, NOPROC		
RESTR1, RESTR2	Point Type AM Custom, Network Point (CLM_nn)		
	Restart CLMI #1, Restart CLMI #2		
	Source NETCDS.CL	Default OFF	Access Lock Operator
	Type Logical		
	Range OFF, ON		

Continued on next page

2.4 Parameter Name Listing, Continued

S1CUROUT

Point Type AM Switch

Current output calculated from OPy and STx_OPy.

Source DIG3CDS.CL **Default** \$ord0 **Access Lock** Engineer
DIG4CDS.CL
DIG5CDS.CL

Type \$2ords – \$5ords (Matches the S1NSTATE, number of states)

Range \$ord0 .. \$ord4

S1CURSTS

Point Type AM Switch

Initial or Current operating state of Switch 1

Source AM **Default** \$ord0 **Access Lock** Program

Type \$2ords – \$5ords (Matches the S1NSTATE, number of states)

Range \$ord0 .. \$ord4

S1NSTATE

Point Type AM Switch

Number of valid states for Switch 1

Source AM **Default** 2 **Access Lock** DEB

Type Integer

Range 2 through 5

S1REQOLD

Point Type AM Switch

Previous requested output

Source DIG3CDS.CL **Default** \$ord0 **Access Lock** Engineer
DIG4CDS.CL
DIG5CDS.CL

Type \$2ords – \$5ords (Matches the S1NSTATE, number of states)

Range \$ord0 .. \$ord4

Continued on next page

2.4 Parameter Name Listing, Continued

S1REQSTS

Point Type AM Switch

Requested state for Switch 1

Source AM **Default** \$ord0 **Access Lock** Operator
(set by S1ACCLVL)

Type \$2ords – \$5ords (Matches the S1NSTATE, number of states)

Range \$ord0 .. \$ord4

S1STATES(0),
S1STATES(1),
S1STATES(2),
S1STATES(3),
S1STATES(4)

Point Type AM Switch

State names for CL Switch states

Source AM **Default** **Access Lock** Engineer-DEB

Type ARRAY (0..4) of String-8

Range Up to eight characters each. Must not be blank.

SALMDSC1,
SALMDSC2,
SALMDSC3

Point Type AM Switch

Switch alarm 1 to 3 descriptions

Source AM **Default** spaces **Access Lock** Engineer

Type String-8

SALMFL1,
SALMFL2,
SALMFL3

Point Type AM Switch

Switch alarm flags 1 to 3

Source **Default** OFF **Access Lock** View

Type Logical

Range OFF, ON

Continued on next page

2.4 Parameter Name Listing, Continued

SCHDCD1,
SCHDCD2,
SCHDCD3,
SCHDCD4

Point Type AM Custom, Network Point (CLM_nn)

Buffer # (1–4) new schedule code

Source NETCDS1.CL (SCHDCD1) **Default** 0 **Access Lock** Operator
NETCDS2.CL (SCHDCD2)
NETCDS3.CL (SCHDCD3)
NETCDS4.CL (SCHDCD4)

Type Real

Range >= 0.0

SCHDIN1,
SCHDIN2,
SCHDIN3,
SCHDIN4

Point Type AM Custom, Network Point (CLM_nn)

Buffer # (1–4) schedule address index #

Source NETCDS1.CL (SCHDIN1) **Default** 0 **Access Lock** Operator
NETCDS2.CL (SCHDIN2)
NETCDS3.CL (SCHDIN3)
NETCDS4.CL (SCHDIN4)

Type Real

Range >= 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

SCOMMAND

Point Type AM Regulatory

Allows the user to do database transfers between the STI point, the Bristol Babcock RTU, and the Smart Transmitter, and to calibrate the transmitter.

Source B_HWSTI.CL **Default** 0 **Access Lock** Operator

Type Real

Range

0	Process mode	(NONE)
1	Download transmitter parameters.....	(DNLOADDB)
2	Upload transmitter parameters	(UPLOADDB)
3	Set the Lower Range value	(SET_LRV)
4	Set the Upper Range value	(SET_URV)
5	Correct the Lower Range value	(COR_LRV)
6	Correct the Upper Range value	(COR_URV)
7	Correct the zero point for the PV value	(COR_INPT)
8	Set calibration parameters to default values.....	(RST_COR)
9	Configure mode	
10	Process mode, Inhibit configuration signals	(procmode)
11	Configure mode, Enable configuration signals ..	(confmode)
99	Test mode.....	(testmode)

Notes regarding transmitter commands:

When transmitter command is specified, the PV and SECVAR signal values are set to NaN, and Questionable Data status is set for both signals.

- While a command is in progress, changes to COMMAND value are ignored.
- At the completion of a command, the same command cannot be issued (until there is a change in COMMAND value).
- If the same command is desired, change the COMMAND value to Configure Mode (CONFMODE), then back to the command value.
- Parameter values are checked for mismatch at the completion of a command.

SCOMMAND

Point Type AM Switch

Future CBO module operation control

Source B_CBO.CL **Default** NaN **Access Lock** Engineer,

Continued on next page

2.4 Parameter Name Listing, Continued

SCRCHPD

Point Type AM Regulatory

Transmitter Scratch Pad: Displays the contents of the Smart Transmitter's scratch pad. It is equivalent to the first part of the S1 parameter in a TDC 3000^X System.

Source B_HWSTI.CL **Default** 0 **Access Lock** Engineer,
CDS page
(class general)

Type String-32

SECVAR

Point Type AM Regulatory

Secondary Variable (read only). Displays the value of the secondary variable of the Smart Transmitter as follows:

Pressure transmitter Temperature of the transmitter
Temperature transmitter..... Cold junction temperature
Flow transmitter Totalized value

Source B_HWSTI.CL **Default** NaN **Access Lock** Engineer

Type Real

Range >= 0.0

SENSRTYP

Point Type AM Regulatory

Smart Transmitter sensor type: When the RTU database and the Smart Transmitter databases have different values, MISMATCH(2) will be ON.

Source B_HWSTI.CL **Default** 0 **Access Lock** Engineer,
build visible

Type Real

Range 8 to 12

Definition of STI SENSRTYP Signal Codes:
8 ST3000 differential pressure (SPT_DP)
9 ST3000 gauge pressure (SPT_GP)
10 ST3000 absolute pressure (SPT_AP)
11 ST3000 temperature (STT)
12 MagneW3000 magnetic flow (SFM)

SERIALNO

Point Type AM Regulatory

Serial number of the transmitter

Source B_HWSTI.CL **Default** 0 **Access Lock** Engineer,
CDS page (class general)

Type String-8

Continued on next page

2.4 Parameter Name Listing, Continued

SERROR	Point Type AM Regulatory			
	Controller error value			
	Source	B_PID3TM.CL	Default 0	Access Lock Engineer
	Type	Real		
	Range	>= 0.0		
SETPOINT	Point Type AM Regulatory			
	Setpoint			
	Source	B_PID3TM.CL	Default 0	Access Lock Engineer
	Type	Real		
	Range	Any real number		
SG	Point Type AM Regulatory			
	Specific gravity of flowing gas			
	Source	B_AGA3.CL B_AGA5.CL B_AGA7.CL B_FPV.CL	Default 0	Access Lock Engineer
	Type	Real		
	Range	>= 0.0		
SMODE	Point Type AM Switch			
	This mode of this module.			
	Source	B_CBO.CL	Default NaN	Access Lock Engineer
	Type	Real		
	Range	0-4		
		1 auto, outputs are reset on error		
		2 auto, no retry outputs are reset on error		
		3 auto, errors are reported without resetting outputs		
		4 auto, no errors are reported without resetting outputs		

Continued on next page

2.4 Parameter Name Listing, Continued

SP	Point Type AM Regulatory Setpoint in Engineering Units Source AM Default 0 Access Lock Operator Type Real Range SPLOLM to SPHILM
SPAN	Point Type AM Regulatory Span of input or output Source B_ANIN.CL Default NaN Access Lock Engineer B_ANOUT.CL B_PID3TM.CL Type Real Range > 0.0
SPHILM	Point Type AM Regulatory Setpoint high limit in Engineering Units of PV Source AM Default NaN Access Lock Supervisor Type Real Range SPLOLM to SPEXEUHI
SPLOLM	Point Type AM Regulatory Setpoint low limit in Engineering Units of PV Source AM Default NaN Access Lock Supervisor Type Real Range SPEXEULO to SPHILM

Continued on next page

2.4 Parameter Name Listing, Continued

SPOINT

Point Type AM Regulatory

Pipe or flange tap construction code

Source B_AGA3.CL **Default** 0 **Access Lock** Engineer

Type Real

Range 1 to 4

Static Pressure
Measured

Pipe
Taps

Flange
Taps

Upstream

4

2

Downstream

3

1

SPOINT

Point Type AM Switch

Field wiring terminal number. Transfer of this data out of the RTU is not supported by BSAP. The user may enter a value here for operator reference.

Source B_CBO.CL **Default** NaN **Access Lock** Engineer

Type Real

Range Any real number, NaN

ST0_OP1,
ST0_OP2,
ST0_OP3

Point Type AM Switch

State 0, Outputs 1 through 3. Defines the logical state (ON or OFF) that is to be written to output # 1, 2, and 3 when the requested state is state 0.

Source DIG3CDS.CL **Default** OFF **Access Lock** Engineer
DIG4CDS.CL
DIG5CDS.CL

Type Logical

Range OFF, ON

Continued on next page

2.4 Parameter Name Listing, Continued

ST1_OP1, ST1_OP2, ST1_OP3	Point Type AM Switch State 1, outputs 1 through 3. Defines the logical state (ON or OFF) that is to be written to output # 1, 2, and 3 when the requested state is state 1. Source DIG3CDS.CL Default OFF Access Lock Engineer DIG4CDS.CL DIG5CDS.CL Type Logical Range OFF, ON
ST2_OP1, ST2_OP2, ST2_OP3	Point Type AM Switch State 2, outputs 1 through 3. Defines the logical state (ON or OFF) that is to be written to output # 1, 2, and 3 when the requested state is state 2. Source DIG3CDS.CL Default OFF Access Lock Engineer DIG4CDS.CL DIG5CDS.CL Type Logical Range OFF, ON
STACDE()	Point Type AM Custom, Network Point (CLM_nn) Buffer station code Source NETCDS.CL Default 0 Access Lock Operator Type Real ARRAY (1..4) Range >= 0.0
START	Point Type AM Switch Command disagree timer running Source DIG3CDS.CL Default OFF Access Lock Engineer DIG4CDS.CL DIG5CDS.CL Type Logical Range OFF, ON

Continued on next page

2.4 Parameter Name Listing, Continued

STATIN1,
STATIN2,
STATIN3,
STATIN4

Point Type AM Custom, Network Point (CLM_nn)

Buffer # (1–4) state address index #

Source NETCDS1.CL (STATIN1) **Default** 0 **Access Lock** Operator
NETCDS2.CL (STATIN2)
NETCDS3.CL (STATIN3)
NETCDS4.CL (STATIN4)

Type Real

Range ≥ 0.0

STATPRES

Point Type AM Regulatory

Static pressure of flowing gas, PSIG

Source B_AGA3.CL **Default** 0 **Access Lock** Engineer
B_FPV.CL

Type Real

Range ≥ 0.0

STATUS

Point Type AM Custom, Builder and Network Points (BUILD_nn and CLM_nn)

Builder Point/Process Status (BUILDCDS)
Status Condition (NETCDS)

Source BUILDCDS.CL **Default** 0 **Access Lock** Engineer
NETCDS.CL

Type Real

Range ≥ 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

STATUS

Point Type AM Regulatory

Status Code. Indicates the current state of the HWSTI module, including the operating mode, command status, and error conditions. Valid status codes are listed below.

Source B_HWSTI.CL **Default** 0 **Access Lock** Engineer

Type Real

Range 0 – 99

Definition of Bristol Babcock STATUS Signal Codes:

0	PROCMODE	Process mode
1	PROCMODE	Process mode with parameter(s) mismatch
2	CONFMODE	Configure mode
3	DNLDEROR	Download completed with rejected parameter
4	DOWNLOAD	Download command in progress
5	DNLD OK	Download completed OK
6	DNLDFAIL	Download failed
7	UPLOAD	Upload command in progress
8	LOADCOMP	Upload completed OK
9	LOADFAIL	Upload failed
10	SET_LRV	Set Lower Range command in progress
11	S_LRV OK	Set Lower Range completed OK
12	SLRVFAIL	Set Lower Range failed
13	SET_URV	Set Upper Range command in progress
14	S_URV OK	Set Upper Range completed OK
15	SURVFAIL	Set Upper Range failed
16	COR_LRV	Correct Lower Range command in progress
17	C_LRV OK	Correct Lower Range completed OK
18	CLRVFAIL	Correct Lower Range failed
19	COR_URV	Correct Upper Range command in progress
20	C_URV OK	Correct Upper Range completed OK
21	CURVFAIL	Correct Upper Range failed
22	COR_ZERO	Correct Zero Point command in progress
23	CZERO OK	Correct Zero Point completed OK
24	ZEROFAIL	Correct Zero Point failed
25	CALIB	Set Calibration Defaults command in progress
26	CALCOMP	Set Calibration Defaults completed OK
27	CALFAIL	Set Calibration Defaults failed
28	INVALID	Invalid COMMAND value
29	POWRFAIL	Command failure due to a power failure.
30	PVTIMEOUT	Transmitter is not operating (PV timeout).
31	BORDFAIL	Interface (I/O board) hardware failure.
32	INVLDDDEV	Invalid DEVICE and/or CHANNEL specified.
33	TESTMODE	Test mode, imaging transmitter data.

Continued on next page

2.4 Parameter Name Listing, Continued

STATUS

Point Type AM Switch

Provides error indication for the CBO module.

Source B_CBO.CL

Default NaN **Access** Lock Engineer

Type Real

Range -12, -10, -9, -8, -4, -3, 0, 2-5

5 = Module is queued and waiting for access to board
4 = Module completed successfully
3 = Operate command completed waiting on Check
2 = Select Check complete
0 = Module completed successfully
-3 = No card in slot DEVICE selected or wrong card
-4 = Invalid mode selected
-8 = Select DO readback error
-9 = Relay contact feedback error
-10 = Operation cancelled by user
-12 = Invalid terminal configuration
-16 = ????

Continued on next page

2.4 Parameter Name Listing, Continued

STBUFER()

Point Type AM Custom, Network Point (CLM_nn)

Work values

Source NETCDS.CL **Default** 0 **Access Lock** Engineer

Type Real ARRAY (1..20)

Range Any real value

STISWVER

Point Type AM Regulatory

Software revision level of the transmitter

Source B_HWSTI.CL **Default** spaces **Access Lock** Engineer,
CDS page
(class general)

Type String–8

STTAG

Point Type AM Regulatory

Transmitter Tag Name. Tag name is used to identify a transmitter to the system and on displays, reports, and logs. When the RTU database and the Smart Transmitter databases have different values, MISMATCH(1) will be ON.

Source B_HWSTI.CL **Default** spaces **Access Lock** Engineer

Type String–8

Continued on next page

2.4 Parameter Name Listing, Continued

STI_EU

Point Type AM Regulatory

Transmitter engineering units

Source B_HWSTI.CL

Default 0

Access Lock Engineer

Type Real

Range 0 to 30

Definition of STI_EU Signal Codes

0	INH2O	(Pressure Xmtr — inches of water)
1	MMHG	(Pressure Xmtr — millimeters of mercury)
2	PSI	(Pressure Xmtr — pounds per sq. inch)
3	KPA	(Pressure Xmtr — kilopascals)
4	MPA	(Pressure Xmtr — millipascals)
5	MBAR	(Pressure Xmtr — millibars)
6	BAR	(Pressure Xmtr — bars)
7	G_SQCM	(Pressure Xmtr — grams per sq. cm)
8	KG_SQCM	(Pressure Xmtr — kilograms per sq. cm)
9	MMH2O	(Pressure Xmtr — millimeters of water)
10	INHG	(Pressure Xmtr — inches of mercury)
11	DEG_C	(Temperature Xmtr — Centigrade)
12	DEG_F	(Temperature Xmtr — Fahrenheit)
13	DEG_K	(Temperature Xmtr — Kelvin)
14	DEG_R	(Temperature Xmtr — Rankine)
15	MV	(Temperature Xmtr — millivolts)
16	V	(Temperature Xmtr — volts)
17	OHMS	(Temperature Xmtr — ohms)
18	CM_HR	(Magnetic flow Xmtr — cubic meters per hour)
19	GAL_HR	(Magnetic flow Xmtr — gallons per hour)
20	LIT_HR	(Magnetic flow Xmtr — liters per hour)
21	CC_HR	(Magnetic flow Xmtr — cubic centimeters per hour)
22	CM_MIN	(Magnetic flow Xmtr — cubic meters per minute)
23	GAL_MIN	(Magnetic flow Xmtr — gallons per minute)
24	LIT_MIN	(Magnetic flow Xmtr — liters per minute)
25	CC_MIN	(Magnetic flow Xmtr — cubic centimeters per minute)
26	CM_DAY	(Magnetic flow Xmtr — cubic meters per day)
27	GAL_DAY	(Magnetic flow Xmtr — gallons per day)
28	KGAL_DA	(Magnetic flow Xmtr — kilogallons per day)
29	BRL_DAY	(Magnetic flow Xmtr — barrels per day)
30	CM_SEC	(Magnetic flow Xmtr — cubic meters per second)

Note: If the STI_EU value is invalid or not used, STI_EU defaults to the transmitter type base units, provided that SENSRTYP is valid according to the following defaults:

Pressure transmitter.....INH20 (Inches of water)

Temperature transmitterDEG_C (Degrees Centigrade)

Flow transmitter.....CM_HR (Cubic meters of water)

Continued on next page

2.4 Parameter Name Listing, Continued

STRNG1,
STRNG2,
STRNG3,
STRNG4

Point Type AM Custom, Network Point (CLM_nn)

Buffer # (1–4) Strings

Source NETCDS1.CL (STRNG1) **Default** spaces **Access Lock** Operator
NETCDS2.CL (STRNG2)
NETCDS3.CL (STRNG3)
NETCDS4.CL (STRNG4)

Type ARRAY (1..150) of String

STSARR()

Point Type AM Custom, Builder Point (BUILD_nn)

Value Status Array

Source BUILD_CDS.CL **Default** spaces **Access Lock** Engineer

Type ARRAY(1..50) of String

T1

Point Type AM Regulatory

Reset rate in repeats per minute

Source B_PID3TM.CL **Default** 0 **Access Lock** Engineer

Type Real

Range ≥ 0.0

T2

Point Type AM Regulatory

Derivative action on measured variable (minutes)

Source B_PID3TM.CL **Default** 0 **Access Lock** Engineer

Type Real

Range ≥ 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

TIMER

Point Type AM Switch

Command disagree timer countdown

Source DIG3CDS.CL **Default** 0 **Access Lock** Engineer
DIG4CDS.CL
DIG5CDS.CL

Type Real

Range >= 0.0

TRACK

Point Type AM Regulatory

Track, when ON, forces the module output in the RTU to follow the RESET terminal input value.

Source B_PID3TM.CL **Default** OFF **Access Lock** Engineer

Type Logical

Range OFF, ON

UPDNIN1, UPDNIN2, UPDNIN3, UPDNIN4

Point Type AM Custom, Network Point (CLM_nn)

Buffer # (1–4) up down address index #

SourceNETCDS1.CL (UPDNIN1) **Default** spaces **Access Lock** Operator
NETCDS2.CL (UPDNIN2)
NETCDS3.CL (UPDNIN3)
NETCDS4.CL (UPDNIN4)

Type Real

Range >= 0.0

UPRDN1, UPRDN2, UPRDN3, UPRDN4

Point Type AM Custom, Network Point (CLM_nn)

Buffer # (1–4) up down code

SourceNETCDS1.CL (UPRDN1) **Default** 0 **Access Lock** Operator
NETCDS2.CL (UPRDN2)
NETCDS3.CL (UPRDN3)
NETCDS4.CL (UPRDN4)

Type Real

Range 0 to 2, 0 for none, 1 for up, 2 for down

Continued on next page

2.4 Parameter Name Listing, Continued

URL

Point Type AM Regulatory

Upper Range Limit: Indicates the upper range limit of the PV at the Smart Transmitter. This limit is a fixed limit and cannot be changed. Refer to the description of the STI_EU parameter for the URL engineering units. When the RTU database and the Smart Transmitter databases have different values, MISMATCH(12) will be ON.

Source B_HWSTI.CL **Default** 0 **Access Lock** Engineer

Type Real

Range ≥ 0.0

URV

Point Type AM Regulatory

Upper Range Value: Defines the upper end of the operating range for the PV parameter value. Refer to the description of the STI_EU parameter for the URV engineering units. User entry for PVEUHI is the user-entered engineering unit value that corresponds to URV. Valid values for URV must be in the range: $LRV \leq (URV \text{ VALUE}) \leq URL$. A value of NaN is invalid. When the RTU database and the Smart Transmitter databases have different values, MISMATCH(11) will be ON. This parameter is mapped as PVEUHI.

Source B_HWSTI.CL **Default** 0 **Access Lock** Supervisor

Type Real

Range ≥ 0.0

VALARR()

Point Type AM Custom, Builder Point (BUILD_nn)

Value Data Array

Source BUILD_CDS.CL **Default** 0 **Access Lock** Engineer

Type ARRAY(1..50) of String

VERROR

Point Type AM Regulatory

Error in validation of entry

Source BUILD_CDS.CL **Default** OFF **Access Lock** Engineer

Type Logical

Range OFF, ON

Continued on next page

2.4 Parameter Name Listing, Continued

VOLUME	Point Type AM Regulatory Volume of gas, MSCF or flow rate, MSCFH Source B_AGA5.CL Default NaN Access Lock Engineer Type Real Range >= 0.0
VOL_CO	Point Type AM Regulatory Volume % CO in gas Source B_AGA5.CL Default NaN Access Lock Engineer Type Real Range >= 0.0
VOL_CO2	Point Type AM Regulatory Volume % CO ₂ in gas Source B_AGA5.CL Default NaN Access Lock Engineer Type Real Range >= 0.0
VOL_CONV	Point Type AM Regulatory Volume conversion factor Source B_AGA5.CL Default NaN Access Lock Engineer Type Real Range >= 0.0
VOL_H2	Point Type AM Regulatory Volume % H ₂ in gas Source B_AGA5 Default NaN Access Lock Engineer Type Real Range >= 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

VOL_H2O	Point Type AM Regulatory
	Volume % H ₂ O in gas
Source B_AGA5.CL	Default NaN Access Lock Engineer
Type Real	
Range >= 0.0	

VOL_H2S	Point Type AM Regulatory
	Volume % H ₂ S in gas
Source B_AGA5.CL	Default NaN Access Lock Engineer
Type Real	
Range >= 0.0	

VOL_HE	Point Type AM Regulatory
	Volume % He in gas
Source B_AGA5.CL	Default NaN Access Lock Engineer
Type Real	
Range >= 0.0	

VOL_N2	Point Type AM Regulatory
	Volume % N ₂ in gas
Source B_AGA5.CL	Default NaN Access Lock Engineer
Type Real	
Range >= 0.0	

Continued on next page

2.4 Parameter Name Listing, Continued

VOL_O2

Point Type AM Regulatory

Volume % O₂ in gas

Source B_AGA5.CL **Default** NaN **Access Lock** Engineer

Type Real

Range >= 0.0

XMITSTAT

Point Type AM Regulatory

Transmitter Status: Displays the contents of the Smart Transmitter's detailed status. It is equivalent to the second part of the S1 parameter in a TDC 3000^X System.

Note: Only 40 of 64 characters can be displayed.

Source B_HWSTI.CL **Default** spaces **Access Lock** Engineer,
CDS page (class general)

Type String-40

ZERO

Point Type AM Regulatory

Input or output zero signal

Source B_ANIN.CL **Default** NaN **Access Lock** Engineer
 B_ANOUT.CL
 B_PID3TM.CL

Type Real

Range >= 0.0

2.5 Dictionary of File Names

Release file names	The following is a complete listing of the files supplied in the release.
B_AGA3.ZI, B_AGA3.ZT	Templates supporting Bristol Babcock AGA3 ACCOL module for calculating gas volume flow through an orifice plate
B_AGA5.ZI, B_AGA5.ZT	Templates supporting Bristol Babcock AGA5 ACCOL module for calculating gas energy from volume, no calorimeter signal (AGA5 method)
B_AGA7.ZI, B_AGA7.ZT	Templates supporting Bristol Babcock AGA3 ACCOL module for calculating gas volume flow measured by turbine flowmeter, positive displacement meter, or vortex shedding meter (AGA7 method)
B_ANIN.ZI, B_ANIN.ZT	Templates supporting image points using the Bristol Babcock High and Low Level Analog input modules
B_ANOUT.ZI, B_ANOUT.ZT	Templates supporting image points using the Bristol Babcock High and Low Level Analog output modules
B_ARRAYI.ZI, B_ARRAYI.ZT	Templates supporting inputs from Bristol Babcock arrays
B_ARRAYO.ZI, B_ARRAYO.ZT	Templates supporting outputs to Bristol Babcock arrays
B_COUNT.ZI, B_COUNT.ZT	Templates supporting image points using the Bristol Babcock Counter module
B_CLMI.ZC	Prototype CLMI Load Directive File for BSAP
B_DEVICE.ZI, B_DEVICE.ZT	NDT Templates support for the RTU Device level
B_DEVICE.ZS	Status Definition File Templates for Bristol Device

Continued on next page

2.5 Dictionary of File Names, Continued

B_DIG3.ZI, B_DIG3.ZT	AM digital composite point (3 states) using Bristol Babcock Digital I/O
B_DIG4.ZI, B_DIG4.ZT	AM digital composite point (4-states) using Bristol Babcock Digital I/O
B_DIG5.ZI, B_DIG5.ZT	AM digital composite point (5-states) using Bristol Babcock Digital I/O
B_FLAG.ZI, B_FLAG.ZT	Templates to implement a Bristol Babcock logical signal as an AM Flag Point
B_FPV.ZI, B_FPV.ZT	Templates supporting Bristol Babcock FPV ACCOL module for calculating the Supercompressibility Factor of a gas
B_HWSTI.ZI, B_HWSTI.ZT	Templates supporting image points using the Bristol Babcock STI module for Honeywell Smartline transmitters
B_INST.EC	Network Interface Layer and Protocol Installation EC for BSAP
B_NUMER.ZI, B_NUMER.ZT	Templates to implement a Bristol Babcock signal in an AM Numeric Point
B_PID3TM.ZI, B_PID3TM.ZT	Templates to implement 3 mode control algorithm using the PID3TM
B_PORT.ZI, B_PORT.ZT	NDT Template support for the Port level using BSAP
B_PORT.ZS	Status Definition File Templates for Bristol Babcock Port level
BSAP.S	S record file for BSAP Protocol
BUILD_NN.EB	Initial exception build file for the Builder Interface Point (BUILD_nn)
BUILDER.DS	Builder Main Display source
BUILDER0.DS	Builder Overlay Display source

Continued on next page

2.5 Dictionary of File Names, Continued

BUILDER1.DS	Builder Page #1 Overlay Display source
BUILDER2.DS	Builder Page #2 Overlay Display source
BUILDER3.DS	Builder Page #3 Overlay Display source
BUILDER4.DS	Builder Page #4 Overlay Display source
BUILDH01.DS	Builder Help Information Display source
CLM.LO	AM Object Code file for the Platform
CLM.SF	CL Compiler Include file for CLM CL Interface
CLM_NN.EB	Initial exception build file for the Network Interface Point (CLM_nn)
CLMI.S	S Record file for CLMI board resident programs
CLMI.ZI, CLMI.ZT	NDT Template support for the CLMI Interface Board level
CLMI.ZS	Status Definition File Templates for Bristol Babcock CLMI Board level.
CLMINNN.ZC	Prototype CLMI Load Directive File
CLMABORT.DS	Network Display source
CLMDDB.DF	Honeywell Specific Variable Data Area Display source
CLMDETD1.DS – CMMDETD4.DS	Device Detail Status Display source, buffer 1 through buffer 4
CLMGRPD1.DS – CLMGRPD4.DS	Point Interface Table Group (PITG) Display source, buffer 1 through buffer 4
CLMERROR.DS	Invalid Entry to Display (source)

Continued on next page

2.5 Dictionary of File Names, Continued

CLMID1.DS	CLM Status Entry Display Subpicture source
CLMOVLY1.DS – CLMOVLY8.DS	Display Source file for Smart Transmitter Interface overlay segments 1 through 8
CLMPIT1.DS – CLMPIT4	PIT Status Display source, buffer 1 through buffer 4
CLMPITE1.DS – CLMPITE4.DS	PITE Status Display source, buffer 1 through buffer 4
CLMRANDR.DS	Release and Revision Subpicture source
CLMSBUF1.DS – CLMSBUF4.DS	Network Device Status Display source, buffer 1 through buffer 4
CLMSCHEM.DS	CLM Status Entry Display source
CLMSMT00.DS	BSAP Smartline Transmitter Configuration Display Source
CLMSTAT1.DS	CLM Status Entry Display Source
DIG3.EC – DIG5.EC	Unlink/Link CL for 3 through 5 State Digital Points
GENAPPL.EB	The exception build file for application display points
GENENTER.DS, GENDISPA.DS, GENDISPB.DS, GENDISPC.DS	Display files for the application displays
INSTALL.EC	Platform Installation EC File
PIDCL.EC	Unlink/Link CL for PID Image Points
PITG.ZI, PITG.ZT	Template support for PITG
PIT.ZI, PIT.ZT	Template support for PIT

2.6 Summary of CL Data Segments

B_AGA3.CL	CDS Image point parameters for the AGA3 ACCOL module. Parameters are NOT BLD_VISIBLE, require Engineer Access level, and appear on the PV algorithm page.
B_AGA5.CL	CDS Image point parameters for the AGA5 ACCOL module. Parameters are NOT BLD_VISIBLE, require Engineer Access level, and appear on the PV algorithm page.
B_AGA7.CL	CDS Image point parameters for the AGA7 ACCOL module. Parameters are NOT BLD_VISIBLE, require Engineer Access level, and appear on the PV algorithm page.
B_ANIN.CL	CDS Image point parameters for the ANIN ACCOL module. Parameters are NOT BLD_VISIBLE, require Engineer Access level, and appear on the PV algorithm page.
B_ANOUT.CL	CDS Image point parameters for the ANOUT ACCOL module. Parameters are NOT BLD_VISIBLE, require Engineer Access level, and appear on the Control algorithm page.
B_COUNT.CL	CDS parameters for Bristol Babcock COUNT ACCOL Module points. Parameters are NOT BLD_VISIBLE, require Engineer Access level, and appear on the PV algorithm page.
B_DG3CDS.CL	Additional CDS ACCOL Bristol Babcock parameters for 3-state composite digital points. Input, Output, and build visible configuration parameters are in the generic file DIG3CDS.CL. Parameters are NOT BLD_VISIBLE and require Engineer ACCESS level. In switch points, all CDS parameters appear on the CDS page.

Continued on next page

2.6 Summary of CL Data Segments, Continued

B_DG4CDS.CL	Additional CDS ACCOL Bristol Babcock parameters for 4-state composite digital points. Input, Output, and build visible configuration parameters are in the generic file DIG4CDS.CL. Parameters are NOT BLD_VISIBLE and require Engineer ACCESS level. In switch points, all CDS parameters appear on the CDS page.
B_DG5CDS.CL	Additional CDS ACCOL Bristol Babcock parameters for 5-state composite digital points. Input, Output, and build visible configuration parameters are in the generic file DIG5CDS.CL. Parameters are NOT BLD_VISIBLE and require Engineer ACCESS level. In switch points, all CDS parameters appear on the CDS page.
B_FPV.CL	CDS Image point parameters for the FPV ACCOL module. Parameters are NOT BLD_VISIBLE, require Engineer Access level, and appear on the PV algorithm page.
B_HWSTI.CL	CDS Image point parameters for the HWSTI ACCOL module. Parameters are NOT BLD_VISIBLE, require Engineer Access level, and appear on the PV algorithm page. Exceptions on build visibility and access level are noted in the foregoing dictionary.
B_PID3TM.CL	CDS Image point parameters for the Bristol Babcock PID3TM ACCOL module. These parameters are NOT BLD_VISIBLE, require Engineer ACCESS level, and appear on the control algorithm page. Exceptions on build visibility and access level are noted in the foregoing dictionary.
BLD2CDS.CL, BUILDCDS.CL	CDS Blocks for CLM Builder Status Point (BUILD_nn). Provides support for the Builder process.

Continued on next page

2.6 Summary of CL Data Segments, Continued

CLM_DMD.CL	CDS for CLM Platform User CL Interface. Includes parameters CLM_DMD, CLM_OUT, CLM_PPS, CLM_PIT, CLM_PORT and CLM_DEV Parameters. Parameters are NOT BLD_VISIBLE, require Engineer ACCESS level, and appear on the CDS page. After running the builder, you must save this data in the AM checkpoint files.
CLM_DMD2.CL	Replacement for CLM_DMD.CL
CLM_DMD3.CL	Replacement for CLM_DMD2.CL
DIG3CDS.CL DIG5CDS.CL	Image point parameters for Generic 2- to 5-state composite digital I/O points. Parameters are NOT BLD_VISIBLE and require Engineer Access level. Exceptions on build visibility and access level are noted in the foregoing dictionary. In switch points, all CDS parameters appear on the CDS page.

2.7 Summary of CL Programs

DIG3CL.AO

Compiled CL routines for 3-State Digital Composite Switch Points. Included is one Generic Block at the general insertion point: (10) named DIG3CL.

ERR3 validates parameters ALMOPT, NODOPTS, and NODINPTS for BADVAL and correct range. Corrects FBTIME if BADVAL. Validates S1NSTATE=3. Invalid configuration is flagged by SALMFL2. Alarms and outputs are blocked by invalid configuration. Parameters with BADVAL are corrected to -1.
MAP3 determines S1CURSTS from inputs and mapping parameters.
ALARM3 off-normal alarming. Uses SALMFL1.
OUTPUT3 Uses output mapping parameters. If NODINPTS = 0, determines S1CURSTS from read back values of OP1, OP2, and OP3. Determines OP1, OP2, and OP3 from S1REQSTS.
Command disagree alarming.
CDALARM3 Command disagree alarm timer.

DIG4CL.AO

Compiled CL routines for 4-State Digital Composite Switch Points. Included is one Generic Block at the general insertion point: (10) named DIG4CL.

ERR4 validates parameters ALMOPT, NODOPTS, and NODINPTS for BADVAL and correct range. Corrects FBTIME if BADVAL. Validates S1NSTATE=4. Invalid configuration is flagged by SALMFL2. Invalid command is flagged by SALMFL3. Alarms and outputs are blocked by invalid configuration. Outputs are blocked by the invalid command alarm. Parameters with BADVAL are corrected to -1.
MAP4 determines S1CURSTS from inputs and mapping parameters.
ALARM4 off-normal alarming. Uses SALMFL1.
OUTPUT4 Uses output mapping parameters. If NODINPTS = 0, determines S1CURSTS from read back values of OP1, OP2, and OP3. Determines OP1, OP2, and OP3 from S1REQSTS.
Command disagree alarming.
CDALARM4 Command disagree alarm timer.

Continued on next page

2.7 Summary of CL Programs, Continued

DIG5CL.AO

Compiled CL routines for 5-State Digital Composite Switch Points. Included is one Generic Block at the general insertion point: (10) named DIG5CL.

ERR5 validates parameters ALMOPT, NODOPTS, and NODINPTS for BADVAL and correct range. Corrects FBTIME if BADVAL. Validates S1NSTATE=5. Invalid configuration is flagged by SALMFL2. Invalid command is flagged by SALMFL3. Alarms and outputs are blocked by invalid configuration. Outputs are blocked by the invalid command alarm. Parameters with BADVAL are corrected to -1.

MAP5 determines S1CURSTS from inputs and mapping parameters.

ALARM5 off-normal alarming. Uses SALMFL1.

OUTPUT5 Uses output mapping parameters. If NODINPTS = 0, determines S1CURSTS from read back values of OP1, OP2, and OP3. Determines OP1, OP2, and OP3 from S1REQSTS. Command disagree alarming.

CDALARM5 Command disagree alarm timer.

Continued on next page

2.7 Summary of CL Programs, Continued

GENRLCDS.CL

This is the Custom Data Segment (CDS) for CLM Application processing. This will support all CLM Applications functionality for picture processes. The application developer can use this CDS to pass values to and from the user. Parameters are not build visible and require Engineer Access Level. In AM Custom points, all parameters appear on the CDS page. It is made up of arrays and will be attached to a special Port LCN point for the application executing on that port.

The point ID will be in the format of "CLMnnPp—" where:

- nn = the CLM node number,
- p = the Application Port number, and
- the – = A, B, and C.

The process requires three separate Custom Points to hold all of the values involved. The CDS is the same for each point but some values will only be used from the point with the "A" code identifier.

PIDCL.AO

CL Program for use with PID image point and Bristol Babcock PID3™ ACCOL Module. Included are the following Blocks at the insertion points indicated:

BLOCK SPPID_CL AT PRE_SP (10). This block stores RTU values for SETPOINT into SP

BLOCK CLPID_CL AT ctl_alg(1). This block performs the CL control algorithm

NETCDS.CL

CDS Block for CLM Network Status Point for CLM/NETWORK PROCESSING. This is the Custom Data Segment (CDS) for CLM processing. This will support all CLM Network functionality for the Network level. Other CDSs are required for complete functionality.

NETCDS1.CL – NETCDS4.CL

CLM Network Status Point CDS for buffer 1 through 4.

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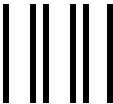
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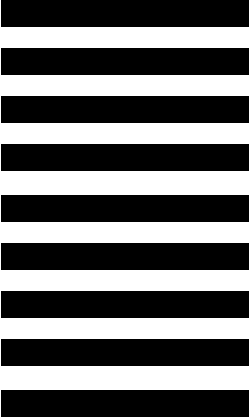
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